

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java

Module - II

Practical – IX

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9. Implement the following JPA applications.

a. Develop a simple Inventory Application Using JPA.

CheckInventory.jsp

```
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.persistence.Persistence"%>
<%@page import="javax.persistence.EntityManager"%>
<%@page import="javax.persistence.EntityManagerFactory"%>
<%@page import="javax.persistence.EntityTransaction"%>
<%@page import="myAPP.Inventory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%
```

```
EntityManagerFactory eMF =
```

```
    Persistence.createEntityManagerFactory("INVJPAPU");
```

```
EntityManager eM = eMF.createEntityManager();
```

```
EntityTransaction eT = null;
```

```
List<Inventory> inventoryList;
```

```
String submit = request.getParameter("AddToInventory");
```

```
if (submit != null && submit.equals("AddToInventory")) {
```

```
    try {
```

```
        String itemname =
```

```
            request.getParameter("itemname");
```

```
        String description =
```

```
            request.getParameter("description");
```

```
        String quantity1 =
```

```
            request.getParameter("quantity");
```

```
        int quantity =
```

```
            Integer.parseInt(quantity1);
```

```
        String addedDate =
```

```
            new java.util.Date().toString();
```

```
        Inventory inv = new Inventory();
```

```
        inv.setItemName(itemname);
```

```
        inv.setDescription(description);
```

```
        inv.setQuantity(quantity);
```

```
        inv.setAddedDate(addedDate);
```

```
        eT = eM.getTransaction();
```

```
        eT.begin();
```

```
        eM.persist(inv);
```

```
        eT.commit();
```

```
    } catch (RuntimeException e) {
```

```
        if (eT != null && eT.isActive()) {
```

```
            eT.rollback(); }
```

```
        throw e; }}
```

```
inventoryList =
```

```
    eM.createQuery("SELECT i FROM Inventory i",
```

```
        Inventory.class)
```

```
        .getResultList();
```

```
eM.close();%>
```

```
<!DOCTYPE html>
```

```

<html>
<head>
  <meta charset="UTF-8">
  <title>Inventory</title>
</head>
<body>
<table border="1">
<tr>
  <td colspan="5" align="center">
    <b>
      Click
      <a href="index.jsp">here</a>
      to add items in the Inventory.
    </b>
  </td>
</tr>
<tr>
  <th>Item ID</th>
  <th>Item Name</th>
  <th>Description</th>
  <th>Quantity</th>
  <th>Added Date & Time</th>
</tr><%
Iterator<Inventory> iterator =
  inventoryList.iterator();
while (iterator.hasNext()) {
  Inventory obj =
    iterator.next();%>
<tr>
  <td><%=obj.getItemID()%></td>
  <td><%=obj.getItemName()%></td>
  <td><%=obj.getDescription()%></td>
  <td><%=obj.getQuantity()%></td>
  <td><%=obj.getAddedDate()%></td>
</tr><%}%>
</table>
</body>
</html>

```

Index.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr>
  <td colspan="2" align="center">
    <b>Add the item in Inventory</b>
  </td>
</tr>
<tr>
  <td>Item Name:</td>
  <td>
    <input name="itemname" maxlength="25" size="50">
  </td>
</tr>
<tr>
  <td>Description:</td>

```

```

<td>
  <textarea rows="5" cols="36"
    name="description"></textarea>
</td>
</tr>
<tr>
  <td>Quantity:</td>
  <td>
    <input name="quantity" maxlength="4" size="5">
  </td>
</tr>
<tr>
  <td colspan="2">
    <input type="submit"
      name="AddToInventory"
      value="AddToInventory">
  </td>
</tr>
</table>
</form>
</body>
</html>

```

Inventory.jsp

```

package myAPP;
import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
@Entity
@Table(name = "inventory")
public class Inventory { @Id
  @GeneratedValue(strategy = GenerationType.IDENTITY)
  @Column(name = "itemid", unique = true, updatable = false)
  private Integer itemid;
  @Column(name = "itemName")
  private String itemName;
  @Column(name = "description")
  private String description;
  @Column(name = "quantity")
  private Integer quantity;
  @Column(name = "addedDate")
  private String addedDate;
  public Inventory() { }
  public Integer getItemID() {
    return itemid;
  }
  public void setItemID(Integer itemid) {
    this.itemid = itemid;
  }
  public String getItemName() {
    return itemName;
  }
  public void setItemName(String itemName) {
    this.itemName = itemName;
  }
  public String getDescription() {
    return description;
  }
  public void setDescription(String description) {
    this.description = description;
  }
  public Integer getQuantity() {
    return quantity;
  }
  public void setQuantity(Integer quantity) {
    this.quantity = quantity;
  }
  public String getAddedDate() {

```

```

return addedDate;}
public void setAddedDate(String addedDate) {
    this.addedDate = addedDate;}}

```

OUTPUT:

The screenshots illustrate the setup of a database connection in NetBeans and the resulting web application output.

Database Configuration Steps:

- New Persistence Unit:** Persistence Unit Name: INVJPAAPU2. Persistence Provider: EclipseLink (JPA 2.1)(default).
- Create Data Source:** JNDI Name: jdbc/inventory. Database Connection: jdbc:derby://localhost:1527/sample [app on APP].
- New Connection Wizard (Locate Driver):** Driver: MySQL (Connector/J driver).
- New Connection Wizard (Customize Connection):**
 - Driver Name: MySQL (Connector/J driver)
 - Host: localhost, Port: 3306
 - Database: Inventory
 - User Name: root
 - Password: [masked]
 - Remember password: ☐
 - Test Connection: [button]
 - JDBC URL: jdbc:mysql://localhost:3306/Inventory?zeroDateTimeBehavior=CONVERT_TO_NULL
 - Connection Succeeded: [message]

Web Application Output:

Add the item in Inventory

Item Name:

Description:

Quantity:

Click [here](#) to add items in the Inventory.

Item ID	Item Name	Description	Quantity	Added Date & Time
1	laptop	Dell laptop	10	Fri Sep 11 13:46:22 IST 2026
2	laptop	Dell laptop	10	Fri Sep 11 13:47:06 IST 2026